Passive-Current Ambiguity in Nonmagnetic Equilibrium Reconstruction: A Controlled Numerical Study

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Abstract

An equilibrium estimator can only recover geometry that its inputs distinguish. We construct and independently check two synthetic free-boundary equilibria with identical measured active currents, total plasma current, geometry and midplane Thomson samples, but opposite unmeasured passive-current states. Their independently located magnetic axes differ by 3.5437 cm. A frozen extension contains ten separately solved equilibria under five known symmetry-breaking drives. Independent source, field and profile checks admit all ten numerical states. Applying classical two-hypothesis Gaussian formulas gives a midplane axis-height minimax RMSE lower bound of 1.7719 cm for the symmetric experiment under a declared synthetic noise model. Moving a diagnostic plane can either improve or worsen distinguishability; adding independent observations cannot reduce information. The construction is an instantaneous, conditional model with unknown passive initial state. It is neither a continuum-certified impossibility theorem nor a demonstrated reconstruction advance on DIII-D or MAST.

1 Question, model and prior work

The scientific target is reconstruction of an axisymmetric poloidal flux $\psi(R, Z)$ from active-coil currents, total plasma current I_p, electron Thomson profiles and machine geometry, without magnetic probes or flux loops. Unmeasured vessel currents and unknown source profiles complicate this inverse problem. This study asks a narrower, falsifiable question: can separately computed, numerically consistent states have substantially different axis heights while their declared instantaneous observations are indistinguishable? It supplies a forward fixture for testing inverse methods, not an inverse method.

Nonuniqueness itself is established prior work. Pentland et al. find multiple static free-boundary equilibria in realistic MAST-U geometry using deflated continuation [1]. Their result concerns the forward GS problem and is not replaced by our simpler construction, which changes an unmeasured external-current state. Up/down reflection and the Gaussian likelihood-ratio calculation below are familiar mathematics. No originality claim attaches to them. Zaitsev et al. also investigated ambiguous current/safety-factor reconstructions and additional-diagnostic accuracy requirements using magnetic measurements [2]. An original reconstruction contribution would need an independently useful algorithm and competitive evidence beyond these observations.

We use the per-radian SI convention

$$\Delta^*\psi = \psi_{RR} - R^{-1}\psi_R + \psi_{ZZ} = -\mu_0 R j_\phi, \qquad B_R = -R^{-1}\psi_Z, \quad B_Z = R^{-1}\psi_R. \tag{1}$$

The field response of current mass $j_\phi\, dR\, dZ$ uses the axisymmetric ring Green function. A finite conductor uses normalized area quadrature multiplied by total ampere-turns. There is no additional $2\pi R$ current multiplier.

2 Frozen synthetic construction and independent checks

The generic device is not MAST hardware. Two active finite-block pairs are centered at $(1, \pm 1.1)$ and $(1.75, \pm 0.6)$ m, each with a 0.06×0.06 m cross-section. Each physical upper/lower block carries 87590.843268 A in the first pair or -47683.612640 A in the second; these are not sums over a pair. Two unmeasured closed rings are centered at $(1.7, \pm 0.8)$ m with 0.04×0.04 m cross-sections. The two states have ring currents $(+1000, -1000)$ or $(-1000, +1000)$ A. Only these hidden currents change in the base pair.

Both states have $I_p = 100$ kA and vacuum $F = RB_\phi = 1$ Tm. With $s = (\psi - \psi_a)/(\psi_b - \psi_a)$, the source family uses FreeGS `ConstrainPaxisIp`, $\alpha_m = 1$, $\alpha_n = 2$, and

$$p(s) = 1000(1-s)^3 \text{ Pa}, \quad n_e(s) = 2 \times 10^{19}(0.2 + 0.8\sqrt{1-s}) \text{ m}^{-3}, \quad T_e(s) = \frac{0.5p(s)}{en_e(s)} \text{ eV}. \quad (2)$$

Here $0 \le s \le 1$ inside the connected plasma. The electron pressure is half the total; the model does not infer ion pressure from measured electrons. Pressure is positive in the interior and zero at the edge, and $F^2 > 0$. Twelve fixed radial channels span $R = 1.02$–1.48 m.

FreeGS is pinned to Git revision `5b41fe5565ff8077bf8139051c80c69416fc9b86` and acquired as library source without examples or equilibrium files [3]. Both members are independently solved at 33^2 and 65^2 resolution on $R \in [0.1, 2]$, $Z \in [-1.5, 1.5]$ m. No reflected bitmap substitutes for a solve. Finite-block quadrature nodes are exactly mirrored between physical upper and lower blocks. An earlier independent triangulation introduced artificial odd quadrature error near conductors; those original runs are retained separately.

A second implementation imports neither the FreeGS solver nor its critical-point routine. It checks fourth-order interior GS differences, the current integral, the declared source law, fields from independently implemented Green responses, and critical-point Hessians. On the base pair the $33 \to 65$ maximum axis-height change is about 0.483 mm, much smaller than the 35.44 mm state separation. This is refinement evidence, not a rigorous continuum error enclosure.

The frozen extension changes the *known* upper/lower PF1 currents to $I_1 + \delta$ and $I_1 - \delta$, with $\delta \in \{0, 100, -100, 300, -300\}$ A, retaining both hidden states in each case. All ten 65^2 states pass the implemented independent numerical gates. The largest current-integral error is 0.0072%, core relative GS residual is 1.98×10^{-6}, and relative core Green-field error is 0.3363%. The final source is recomputed from each converged field.

Review found that the original protocol also requested a whole-domain field comparison, whereas the implemented gate tested only the core. A retained retrospective addendum completes that omitted check on the same frozen states, including vacuum and excluding conductor neighborhoods and one outer grid cell. All ten pass the original 2% threshold; the largest whole-domain relative field error is 0.0980%. Both base errors decrease from 0.4628% at 33^2 to 0.0968% at 65^2. Core and whole-domain norms use different regions and field scales; the smaller whole-domain ratio does not imply a stronger core accuracy result. The conductor buffer changes with grid spacing, so this is not a fixed-region convergence study.

Diagnostic queries follow the declared observation operator: interpolate *plasma* flux and add exact quadrature responses of the coils at each sensor. Interpolating the total raster instead changes this operator slightly. Corrected independent mean calculations agree within 2.38×10^{-15} relative error; the stricter original acceptance threshold was retained. Failed checker outputs and both evaluator versions remain available.

3 Conditional information calculation

For each fixed drive and diagnostic plane, let the two mean vectors be $\mu_+, \mu_- \in \mathbb{R}^{24}$, containing twelve temperatures followed by twelve densities. Declare independent Gaussian channels with a *common* covariance $\Sigma = \text{diag}(\sigma_i^2)$. With componentwise midpoint $\bar{\mu} = (\mu_+ + \mu_-)/2$, the frozen

standard deviations are

$$\sigma_{T,i} = 0.05 \max(\bar{T}_i, 5\,\mathrm{eV}), \qquad \sigma_{n,i} = 0.05 \max(\bar{n}_i, 10^{17}\,\mathrm{m}^{-3}). \tag{3}$$

These are synthetic modeling assumptions, not experimentally calibrated diagnostic uncertainties. All sampled channels lie inside the plasma in both members. The frozen protocol retains membership and would assign zero means outside, rather than discard inconvenient channels.

Write $d^2 = (\mu_+ - \mu_-)^\top \Sigma^{-1} (\mu_+ - \mu_-)$ and let $\Delta = |Z_{a,+} - Z_{a,-}|$. The following classical result states exactly what is calculated; equal priors define the binary experiment, not a population model.

Proposition 1. *For the two declared Gaussian observation distributions, the equal-prior optimal classification error is $P_e = \Phi(-d/2)$. For any class containing both states, the minimax axis-height mean squared error is at least*

$$R_2 = \frac{\Delta^2}{4}\, \mathbb{E}_{U\sim N(0,1)} \left[\operatorname{sech}^2\left(\frac{dU + d^2/2}{2} \right) \right]. \tag{4}$$

At $d = 0$, this gives $R_2 = \Delta^2/4$.

Proof. The log likelihood ratio under the positive state is normal with mean $d^2/2$ and variance d^2, so the likelihood-ratio test has error $\Phi(-d/2)$. For squared loss, maximum risk is at least equal-prior average risk. Its minimizer is the posterior mean, whose conditional variance is $\Delta^2 q(1-q)$, where q is the positive-state posterior probability. Since $q(1-q) = \operatorname{sech}^2(L/2)/4$ is even in the log likelihood ratio L, its mixture expectation equals its expectation under either component, giving (4). □

The posterior mean need not be a GS equilibrium. Allowing it strengthens the scope of the lower bound but does not make it a recommended physical output. Ordinary adaptive integration evaluates (4); reported quadrature error estimates are not interval-arithmetic certificates.

4 Results and physical interpretation

The symmetric midplane distributions are indistinguishable to numerical precision. The axis-height RMSE lower bound is 1.7719 cm. Known asymmetry can make the states distinguishable: the midplane binary error falls from 0.5 to about 0.329 at $\delta = \pm 100$ A and 0.0914 at $\delta = \pm 300$ A. Neither exact symmetry nor universal ambiguity is assumed in the perturbed experiments; every frozen case is reported.

Replacing the plane and adding observations are different experiments. At $\delta = -300$ A, moving from $Z = 0$ to $Z = 1$ cm *increases* binary error from 0.0914 to 0.2033. Adding that independent plane instead gives 0.0583. For independent blocks the squared whitened separations add, so the second result follows from ordinary information accumulation, not a new design theorem. These results support retaining actual diagnostic geometry and explicit nuisance states when evaluating an inverse method.

A separate real-data *oracle-source* audit checks the response machinery on 48 training frames from 12 DIII-D discharges. Source current and plasma support are extracted from reference flux, and the supplied signed F-coil response is added without a fitted gain. Median relative field errors are 1.6848% in the interior and 3.1677% in the evaluated vacuum region. These checks use the answer to construct the source; they establish conditional representation consistency, not predictive accuracy. Missing E-coil information, passive sources and flux gauge remain. The original negative-study manuscript reports the separate baseline and rejected inverse candidates.

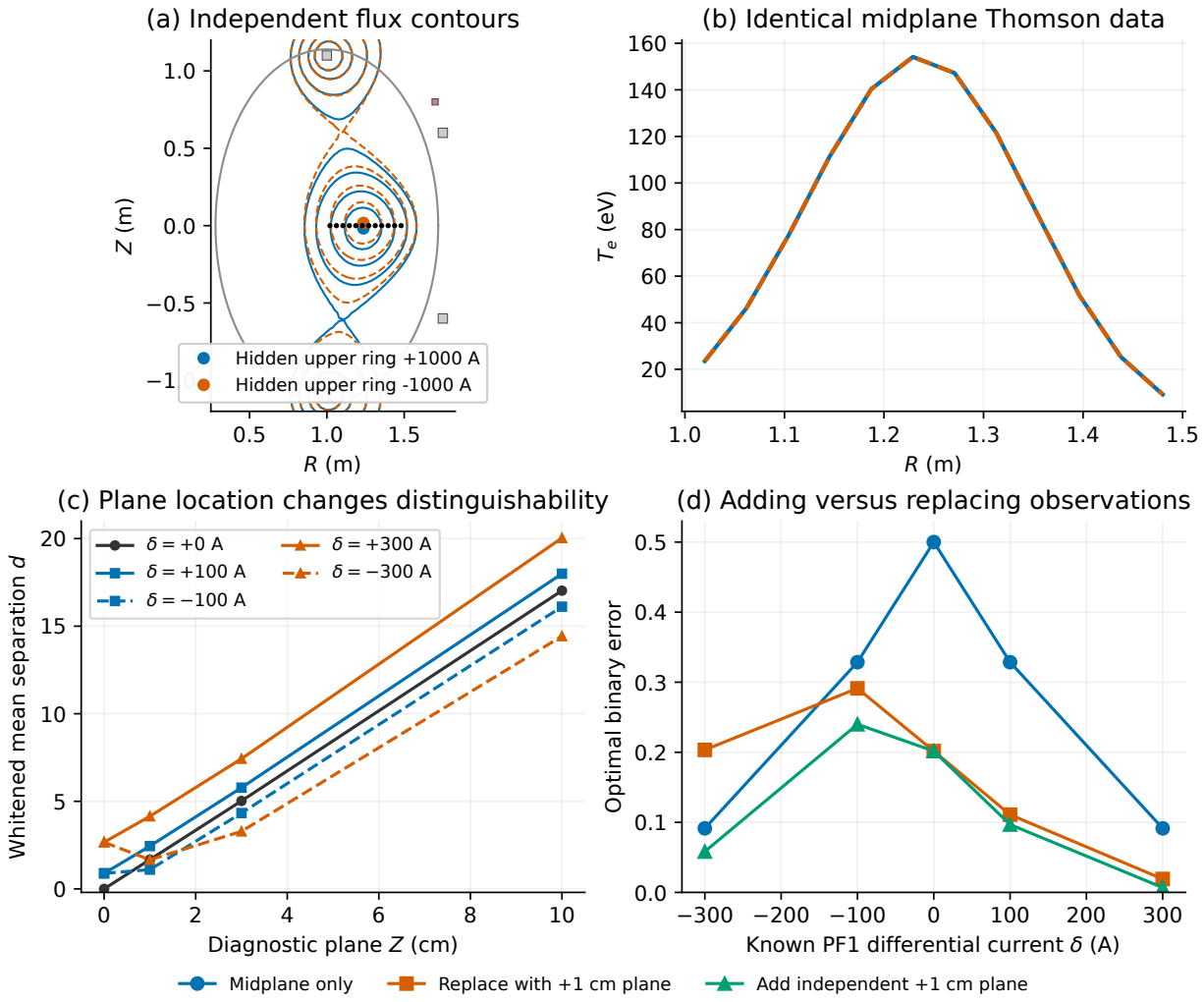


Figure 1: Controlled synthetic observation experiments, generated from retained independent checks. (a) Normalized flux contours at $s = 0.2, 0.5, 0.8, 1$; outer contour branches are not additional confined plasmas. Gray blocks are measured active coils; magenta blocks are hidden rings. (b) Base-pair midplane temperatures overlap; densities agree as well. The axis gap is 3.5437 cm. (c) Whitened separation for each replacement diagnostic plane and known drive. (d) Binary error for a single plane or two independent planes. Lines join discrete frozen cases, not a fitted continuum law. The Gaussian covariance is synthetic and assumes independent channels.

δ (A)	Axis gap (cm)	$P_e(0)$	$P_e(1\,\text{cm})$	$P_e(3\,\text{cm})$	Floor at 0 (cm)
-300	3.5569	0.0914	0.2033	0.05032	0.9190
-100	3.5453	0.3286	0.2912	0.01521	1.6179
+0	3.5437	0.5000	0.2017	0.00603	1.7719
+100	3.5453	0.3286	0.1109	0.00193	1.6179
+300	3.5569	0.0914	0.0191	0.00010	0.9190

Table 1: Independent numerical two-state calculations. P_e is optimal binary classification error for the plane named in parentheses. "Floor" is the square root of (4) for the midplane observation model, expressed in cm. It is conditional on the declared numerical states and noise model.

5 Limits and the remaining research requirement

An instantaneous unknown passive initial state is essential to this witness. A declared copper-ring model has positive inductance eigenvalues, energy about 9.02 J and odd-mode L/R time about 79.5 ms. This does not prove a coupled plasma/circuit trajectory, preparation under an identical known history, permanent unpowered current or vertical stability. Known initial states, informative histories, additional sensors or off-plane profiles can remove this particular ambiguity. Reflection-even scalars are not separated here.

Numerical states are not certified continuum solutions. If validated output errors were e_+, e_- and whitened mean errors were η_+, η_-, a conservative propagation would use $\Delta_{\text{lower}} = \max(0, \widehat{\Delta} - e_+ - e_-)$ and $d_{\text{upper}} = \widehat{d} + \eta_+ + \eta_-$ with fixed covariance. Such enclosures and a rigorous integration bound have not been computed. The word "certified" therefore does not describe the physical error floor.

These fixtures and classical bounds do not supply a new inverse algorithm. Competitive source-to-target reconstruction, credible model-error coverage and a defensible novelty claim remain open. No target labels or blind evaluation were used.

Reproducibility and review. The frozen protocol, all ten forward packets, failed intermediates, independent checker, information-bound tests and figure/table generator are retained in the accompanying repository. Exact source and input hashes bind the principal calculations. A separate numerical reviewer rebuilt the sensor means using the opposite spline orientation and a direct elliptic kernel, confirming the covariance, distance and pooled-information arithmetic to roundoff. These checks use a separately implemented numerical calculation. Reproduction commands and original source licenses are documented in `REPRODUCE.md`.

References

[1] K. Pentland, N. C. Amorisco, P. E. Farrell and C. J. Ham, "Multiple solutions to the static forward free-boundary Grad–Shafranov problem on MAST-U," arXiv:2503.05674v5 (2025).

[2] F. S. Zaitsev et al., "Analyses of substantially different plasma current densities and safety factors reconstructed from magnetic diagnostics data," *Nuclear Fusion* **51**, 103044 (2011), doi:10.1088/0029-5515/51/10/103044.

[3] FreeGS developers, *FreeGS*, source revision `5b41fe5565ff8077bf8139051c80c69416fc9b86`, official source repository (accessed 29 October 2026).